

CASOGEOGRAPHICS

TECHNICAL AND DATA APPENDIX

Data model, calculations, audit protocol and reproducibility notes

COMPANION TO THE MAIN RESEARCH REPORT

Electoral geography • GIS • Historical persistence • Neighbourhood strategy

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Document control

Field	Value
Publication	The Long Geography of Politics
Companion	Technical and Data Appendix
Version	1.0 — June 2026
Author	Antonio Caso
Publisher	casogeographics
Primary dataset	Sezioni (2).xlsx
Analytical coverage	19 polling-place aggregates; 189 section records
Status	Methodological pilot; final official-data reconciliation required

Purpose

This appendix documents the working data structure, transformations, calculations, quality checks and ethical constraints required to reproduce or extend the Taranto case study. It separates observed values from interpretation and identifies unresolved data issues.

1. Source inventory and provenance

Source	Format	Use	Status
Sezioni (2).xlsx	Excel workbook	Polling-place, candidate and section-level results	Primary working dataset
Taranto municipal-election geographical essay	PDF	Urban interpretation and hypotheses	Author-produced contextual source
Sezioni amministrative 2025	Google My Maps	Polling-place location and public visualisation	Interactive source
The Long Geography of Politics	DOCX	Methodological and comparative synthesis	Main publication
Global Paleo-Political Laboratory	Web pages	Comparative-case origin and Tree metaphor	Revised and de-determined in report

Public map: [Google My Maps — Sezioni amministrative 2025](#)

2. Workbook architecture

The workbook contains one section registry, one analytical note sheet and 19 polling-place worksheets. Each polling-place worksheet combines first-round candidate totals with section-level runoff data. The workbook is therefore a hybrid analytical file rather than a normalised database.

Component	Rows / sheets	Description
SEZIONI registry	215 rows including header and blanks	Section number, polling place and address; 189 section identifiers used in runoff tables
Elettori 1 turno	57 rows	Analytical notes, selected totals and thematic prompts
Polling-place sheets	19 sheets	First-round candidate summary and runoff section records
Derived JSON	19 aggregates / 189 sections	Normalised structure created for the interactive atlas

3. Normalised data model

Entity	Primary key	Main fields
PollingPlace	polling_place_id	name, address, analytical macro-area, notes
FirstRoundResult	polling_place_id + candidate	votes, reported percentage, normalisation basis
RunoffSectionResult	section_number	polling place, Bitetti votes, Tacente votes, total, percentages
DerivedAggregate	polling_place_id	runoff totals, margins, retention proxy, competitiveness
SpatialRecord	polling_place_id	latitude, longitude, geocoding source, confidence
EvidenceRecord	claim_id	claim, source, method, confidence, caveat

Recommended database design

For future projects, store election, geography and evidence data in separate linked tables. Do not use worksheet names as identifiers; create stable IDs and preserve original labels in a source_name field.

4. Data dictionary

Field	Type	Definition	Quality rule
section_number	Integer	Official or working polling-section identifier	Required; unique in runoff registry
polling_place_name	Text	Institution or building hosting one or more sections	Standardise abbreviations
address	Text	Working address string	Geocode with source and confidence
candidate	Category	Candidate name in the relevant round	Use controlled vocabulary
first_round_votes	Integer	Candidate votes in polling-place aggregate	Verify official basis
first_round_pct	Decimal	Candidate share reported in workbook	Currently normalised within analysed set
runoff_votes	Integer	Candidate votes in one section	Observed working value
runoff_total	Integer	Sum of the two runoff candidates	Derived
runoff_pct	Decimal	Candidate runoff votes / runoff total × 100	Derived
runoff_margin_pp	Decimal	Bitetti share minus Tacente share	Derived percentage points
retention_proxy	Decimal	Runoff total / first-round analysed total	Interpret cautiously
competitiveness	Decimal	100 – absolute runoff margin	Derived index
candidate_specialisation	Decimal	Local share minus city analysed share	Derived
geocode_confidence	Category	Exact / interpolated / locality / unresolved	Required for formal atlas

5. Core calculations

Indicator	Formula	Interpretation
Runoff total	$B_i + T_i$	Votes cast for the two runoff candidates in section or aggregate i
Bitetti runoff share	$100 \times B_i / (B_i + T_i)$	Candidate share within valid two-candidate total
Runoff margin	Bitetti share – Tacente share	Positive values favour Bitetti
Electoral weight	Aggregate valid votes / city valid votes	Share of observed electoral volume
Candidate specialisation	Local candidate share – city candidate share	Local over- or under-performance
Concentration	Σ local candidate votes in selected areas / total candidate votes	Candidate dependence on specific territories
Retention proxy	Runoff total / first-round working total	Not equivalent to individual voter retention
Competitiveness	$100 - \text{runoff margin} $	Higher values indicate a closer aggregate

Ecological-inference warning

A numerical similarity between first-round and runoff totals does not identify individual vote transfers. Transfer language must remain probabilistic unless supported by survey, panel or individual-level data.

6. Dataset-wide summary

Metric	Working value
Polling-place aggregates	19
Runoff section records	189
Section-number span	1–189
First-round analysed volume	84,564
Bitetti first-round analysed votes	33,641
Tacente first-round analysed votes	23,283
Lazzaro first-round analysed votes	17,787
Angolano first-round analysed votes	9,853
Runoff working total	72,250
Bitetti runoff share	54.71%
Tacente runoff share	45.29%

7. Polling-place aggregate register

ID	Polling place	Address	1st rnd*	Runoff	Bitetti %	Sections
1	IST. DE AMICIS C/O EDIF. UGO FOSCOLO	VIA BRUNELLESCHI, 20 TALSANO	7176	5,884	53.0%	14
2	XXIV CIRCOLO DIDATTICO TRAMONTONE	VIA GREGORIO VII - TALSANO	11667	5,735	56.1%	12
3	IST. R. FRASCOLLA LAMA	C/O EDIFICIO DI VIA CICLAMINI - LAMA	3264	2,904	57.2%	5
4	IST. RENATO FRASCOLLA SAN VITO	VIALE JONIO, 513 SAN VITO	2668	2,250	55.4%	5
5	IST. PROF. ARCHIMEDE EX M.COSTA	VIA LAGO DI MISURINA, 22	4740	4,137	57.7%	10
6	IST. MARTELOTTO	VIA SCOGLIO DEL TONNO	3585	3,307	55.2%	6
7	SCUOLA MEDIA ALESSANDRO VOLTA	VIA VENEZIA, 75	6065	5,417	61.0%	14
8	IST. RENATO MORO	VIA UMBRIA, 261	3749	4,150	51.9%	11
9	IST. C.G.VIOLA C/O EDIFICIO C. BATTISTI	VIA ACTON	4320	4,167	55.0%	12
10	IST. SAN GIOVANNI BOSCO	VIA POLIBIO, 46	2575	2,281	53.7%	6
11	IST. DANTE A. C/O EDIF. ACANFORA	INGR. VIA DANTE, VIA OBERDAN	4719	5,268	50.9%	17
12	XII CIRCOLO DIDATTICO L.TEMPESTA C/O EDIFICIO DON MILANI	VIA VENETO, 110	4567	4,054	54.7%	11
13	SCUOLA MEDIA C.COLOMBO	VIA MEDAGLIE D'ORO, 117	4618	4,935	56.5%	14
14	IST. XXV LUGLIO - G.BETTOLO	VIA PUPINO, 4	1761	1,620	52.9%	4
15	IST. XXV LUGLIO - G.BETTOLO	INGRESSO VIA PITAGORA, 73	4589	4,116	52.9%	11
16	LICEO GINNASIO ARISTOSSENO	VIALE VIRGILIO, 15	1466	1,254	54.9%	5
17	IST. GALILEO GALILEI	VICO CARDUCCI, 9	922	715	52.3%	3
18	SCUOLA ELEMENTARE G.B. VICO	INGR.VIA ARCHIMEDE, VIA SS. ANGELI CUSTODI	5439	4,583	49.7%	16
19	ISTITUTO COMPRENSIVO L.PIRANDELLO	EDIFICIO VIA PASTORE,5	6674	5,473	55.5%	13

* First-round working total represented in the aggregate worksheet; verify normalisation and excluded candidates against official returns.

8. Aggregate candidate profiles

ID	Polling place	Bitetti	Tacente	Lazzaro	Angolano	Runoff B
1	IST. DE AMICIS C/O EDIF. UGO FOSC	35.9	23.1	22.6	10.7	53.0
2	XXIV CIRCOLO DIDATTICO TRAMONTONE	40.2	24.8	23.7	11.4	56.1
3	IST. R. FRASCOLLA LAMA	40.8	24.6	22.9	11.7	57.2
4	IST. RENATO FRASCOLLA SAN VITO	38.0	22.1	29.7	10.2	55.4
5	IST. PROF. ARCHIMEDE EX M.COSTA	43.6	25.2	20.2	11.0	57.7
6	IST. MARTELOTTA	38.2	26.5	21.4	13.9	55.2
7	SCUOLA MEDIA ALESSANDRO VOLTA	43.7	22.3	21.6	12.5	61.0
8	IST. RENATO MORO	38.4	26.3	23.9	11.3	51.9
9	IST. C.G.VIOLA C/O EDIFICIO C. BA	41.9	28.0	18.2	11.9	55.0
10	IST. SAN GIOVANNI BOSCO	39.0	30.9	17.8	12.3	53.7
11	IST. DANTE A. C/O EDIF. ACANFORA	36.4	31.5	20.0	12.1	50.9
12	XII CIRCOLO DIDATTICO L.TEMPESTA	40.2	27.3	21.4	11.1	54.7
13	SCUOLA MEDIA C.COLOMBO	41.3	24.9	23.4	10.4	56.5
14	IST. XXV LUGLIO - G.BETTOLO	32.5	29.8	27.2	10.5	52.9
15	IST. XXV LUGLIO - G.BETTOLO	35.1	28.0	23.8	13.1	52.9
16	LICEO GINNASIO ARISTOSSENSO	37.4	27.6	23.1	11.9	54.9
17	IST. GALILEO GALILEI	47.0	34.2	12.2	6.7	52.3
18	SCUOLA ELEMENTARE G.B. VICO	38.1	35.5	13.6	12.8	49.7
19	ISTITUTO COMPRENSIVO L.PIRANDELLO	41.6	35.6	11.8	11.0	55.5

9. Section-level runoff registry

The following register is generated from the 19 polling-place worksheets. It preserves the working values and supports recalculation of all aggregate runoff percentages.

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
1	2	IST. DE AMICIS C/O EDIF. UGO	165	153	318	51.89	48.11
1	3	IST. DE AMICIS C/O EDIF. UGO	159	126	285	55.79	44.21
1	4	IST. DE AMICIS C/O EDIF. UGO	225	154	379	59.37	40.63
1	5	IST. DE AMICIS C/O EDIF. UGO	221	194	415	53.25	46.75
1	6	IST. DE AMICIS C/O EDIF. UGO	198	162	360	55.00	45.00
1	7	IST. DE AMICIS C/O EDIF. UGO	166	152	318	52.20	47.80
1	8	IST. DE AMICIS C/O EDIF. UGO	239	202	441	54.20	45.80
1	9	IST. DE AMICIS C/O EDIF. UGO	245	209	454	53.96	46.04
1	10	IST. DE AMICIS C/O EDIF. UGO	287	243	530	54.15	45.85
1	11	IST. DE AMICIS C/O EDIF. UGO	228	251	479	47.60	52.40
1	12	IST. DE AMICIS C/O EDIF. UGO	229	222	451	50.78	49.22
1	13	IST. DE AMICIS C/O EDIF. UGO	319	288	607	52.55	47.45
1	14	IST. DE AMICIS C/O EDIF. UGO	256	244	500	51.20	48.80
1	15	IST. DE AMICIS C/O EDIF. UGO	184	163	347	53.03	46.97
2	16	XXIV CIRCOLO DIDATTICO TRAMON	276	123	399	69.17	30.83
2	17	XXIV CIRCOLO DIDATTICO TRAMON	260	200	460	56.52	43.48
2	18	XXIV CIRCOLO DIDATTICO TRAMON	137	179	316	43.35	56.65
2	19	XXIV CIRCOLO DIDATTICO TRAMON	223	241	464	48.06	51.94

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
2	20	XXIV CIRCOLO DIDATTICO TRAMON	327	223	550	59.45	40.55
2	21	XXIV CIRCOLO DIDATTICO TRAMON	251	199	450	55.78	44.22
2	22	XXIV CIRCOLO DIDATTICO TRAMON	179	167	346	51.73	48.27
2	23	XXIV CIRCOLO DIDATTICO TRAMON	333	252	585	56.92	43.08
2	34	XXIV CIRCOLO DIDATTICO TRAMON	257	197	454	56.61	43.39
2	35	XXIV CIRCOLO DIDATTICO TRAMON	353	277	630	56.03	43.97
2	36	XXIV CIRCOLO DIDATTICO TRAMON	334	218	552	60.51	39.49
2	37	XXIV CIRCOLO DIDATTICO TRAMON	287	242	529	54.25	45.75
3	24	IST. R. FRASCOLLA LAMA	343	243	586	58.53	41.47
3	25	IST. R. FRASCOLLA LAMA	324	260	584	55.48	44.52
3	26	IST. R. FRASCOLLA LAMA	295	204	499	59.12	40.88
3	27	IST. R. FRASCOLLA LAMA	340	318	658	51.67	48.33
3	28	IST. R. FRASCOLLA LAMA	358	219	577	62.05	37.95
4	29	IST. RENATO FRASCOLLA SAN VIT	273	188	461	59.22	40.78
4	30	IST. RENATO FRASCOLLA SAN VIT	221	204	425	52.00	48.00
4	31	IST. RENATO FRASCOLLA SAN VIT	228	210	438	52.05	47.95
4	32	IST. RENATO FRASCOLLA SAN VIT	235	170	405	58.02	41.98
4	33	IST. RENATO FRASCOLLA SAN VIT	289	232	521	55.47	44.53
5	38	IST. PROF. ARCHIMEDE EX M.COS	235	150	385	61.04	38.96
5	39	IST. PROF. ARCHIMEDE EX M.COS	196	141	337	58.16	41.84
5	40	IST. PROF. ARCHIMEDE EX M.COS	178	157	335	53.13	46.87
5	41	IST. PROF. ARCHIMEDE EX M.COS	206	161	367	56.13	43.87
5	42	IST. PROF. ARCHIMEDE EX M.COS	212	107	319	66.46	33.54
5	47	IST. PROF. ARCHIMEDE EX M.COS	190	138	328	57.93	42.07

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
5	48	IST. PROF. ARCHIMEDE EX M.COS	157	123	280	56.07	43.93
5	49	IST. PROF. ARCHIMEDE EX M.COS	381	300	681	55.95	44.05
5	50	IST. PROF. ARCHIMEDE EX M.COS	277	196	473	58.56	41.44
5	51	IST. PROF. ARCHIMEDE EX M.COS	353	279	632	55.85	44.15
6	52	IST. MARTELLOTTA	276	225	501	55.09	44.91
6	53	IST. MARTELLOTTA	204	220	424	48.11	51.89
6	54	IST. MARTELLOTTA	410	360	770	53.25	46.75
6	55	IST. MARTELLOTTA	290	234	524	55.34	44.66
6	56	IST. MARTELLOTTA	267	190	457	58.42	41.58
6	188	IST. MARTELLOTTA	378	253	631	59.90	40.10
7	57	SCUOLA MEDIA ALESSANDRO VOLTA	274	166	440	62.27	37.73
7	58	SCUOLA MEDIA ALESSANDRO VOLTA	272	171	443	61.40	38.60
7	59	SCUOLA MEDIA ALESSANDRO VOLTA	284	171	455	62.42	37.58
7	60	SCUOLA MEDIA ALESSANDRO VOLTA	221	130	351	62.96	37.04
7	61	SCUOLA MEDIA ALESSANDRO VOLTA	233	188	421	55.34	44.66
7	62	SCUOLA MEDIA ALESSANDRO VOLTA	185	129	314	58.92	41.08
7	63	SCUOLA MEDIA ALESSANDRO VOLTA	211	140	351	60.11	39.89
7	64	SCUOLA MEDIA ALESSANDRO VOLTA	180	141	321	56.07	43.93
7	65	SCUOLA MEDIA ALESSANDRO VOLTA	241	134	375	64.27	35.73
7	66	SCUOLA MEDIA ALESSANDRO VOLTA	186	105	291	63.92	36.08
7	67	SCUOLA MEDIA ALESSANDRO VOLTA	217	119	336	64.58	35.42
7	68	SCUOLA MEDIA ALESSANDRO VOLTA	282	208	490	57.55	42.45
7	69	SCUOLA MEDIA ALESSANDRO VOLTA	224	126	350	64.00	36.00
7	189	SCUOLA MEDIA ALESSANDRO VOLTA	294	185	479	61.38	38.62
8	70	IST. RENATO MORO	197	219	416	47.36	52.64

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
8	71	IST. RENATO MORO	241	190	431	55.92	44.08
8	105	IST. RENATO MORO	237	201	438	54.11	45.89
8	106	IST. RENATO MORO	189	147	336	56.25	43.75
8	107	IST. RENATO MORO	191	185	376	50.80	49.20
8	111	IST. RENATO MORO	189	207	396	47.73	52.27
8	112	IST. RENATO MORO	208	173	381	54.59	45.41
8	113	IST. RENATO MORO	190	173	363	52.34	47.66
8	114	IST. RENATO MORO	160	160	320	50.00	50.00
8	115	IST. RENATO MORO	193	153	346	55.78	44.22
8	116	IST. RENATO MORO	160	187	347	46.11	53.89
9	72	IST. C.G.VIOLA C/O EDIFICIO C	213	141	354	60.17	39.83
9	73	IST. C.G.VIOLA C/O EDIFICIO C	151	135	286	52.80	47.20
9	74	IST. C.G.VIOLA C/O EDIFICIO C	231	142	373	61.93	38.07
9	75	IST. C.G.VIOLA C/O EDIFICIO C	174	181	355	49.01	50.99
9	76	IST. C.G.VIOLA C/O EDIFICIO C	209	156	365	57.26	42.74
9	77	IST. C.G.VIOLA C/O EDIFICIO C	210	155	365	57.53	42.47
9	78	IST. C.G.VIOLA C/O EDIFICIO C	191	168	359	53.20	46.80
9	79	IST. C.G.VIOLA C/O EDIFICIO C	176	150	326	53.99	46.01
9	80	IST. C.G.VIOLA C/O EDIFICIO C	184	154	338	54.44	45.56
9	81	IST. C.G.VIOLA C/O EDIFICIO C	176	143	319	55.17	44.83
9	82	IST. C.G.VIOLA C/O EDIFICIO C	179	160	339	52.80	47.20
9	83	IST. C.G.VIOLA C/O EDIFICIO C	199	189	388	51.29	48.71
10	89	IST. SAN GIOVANNI BOSCO	242	172	414	58.45	41.55
10	90	IST. SAN GIOVANNI BOSCO	217	178	395	54.94	45.06
10	91	IST. SAN GIOVANNI BOSCO	200	194	394	50.76	49.24
10	92	IST. SAN GIOVANNI BOSCO	214	197	411	52.07	47.93
10	93	IST. SAN GIOVANNI BOSCO	167	150	317	52.68	47.32

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
10	94	IST. SAN GIOVANNI BOSCO	186	164	350	53.14	46.86
11	84	IST. DANTE A. C/O EDIF. ACANF	303	228	531	57.06	42.94
11	85	IST. DANTE A. C/O EDIF. ACANF	286	231	517	55.32	44.68
11	88	IST. DANTE A. C/O EDIF. ACANF	109	155	264	41.29	58.71
11	95	IST. DANTE A. C/O EDIF. ACANF	142	107	249	57.03	42.97
11	96	IST. DANTE A. C/O EDIF. ACANF	118	142	260	45.38	54.62
11	97	IST. DANTE A. C/O EDIF. ACANF	129	154	283	45.58	54.42
11	98	IST. DANTE A. C/O EDIF. ACANF	197	218	415	47.47	52.53
11	99	IST. DANTE A. C/O EDIF. ACANF	218	198	416	52.40	47.60
11	100	IST. DANTE A. C/O EDIF. ACANF	170	166	336	50.60	49.40
11	101	IST. DANTE A. C/O EDIF. ACANF	151	126	277	54.51	45.49
11	102	IST. DANTE A. C/O EDIF. ACANF	161	147	308	52.27	47.73
11	103	IST. DANTE A. C/O EDIF. ACANF	107	130	237	45.15	54.85
11	104	IST. DANTE A. C/O EDIF. ACANF	131	148	279	46.95	53.05
11	149	IST. DANTE A. C/O EDIF. ACANF	120	144	264	45.45	54.55
11	150	IST. DANTE A. C/O EDIF. ACANF	139	120	259	53.67	46.33
11	151	IST. DANTE A. C/O EDIF. ACANF	103	90	193	53.37	46.63
11	152	IST. DANTE A. C/O EDIF. ACANF	99	81	180	55.00	45.00
12	43	XII CIRCOLO DIDATTICO L.TEMPE	172	166	338	50.89	49.11
12	44	XII CIRCOLO DIDATTICO L.TEMPE	220	199	419	52.51	47.49
12	45	XII CIRCOLO DIDATTICO L.TEMPE	174	112	286	60.84	39.16
12	46	XII CIRCOLO DIDATTICO L.TEMPE	221	129	350	63.14	36.86
12	108	XII CIRCOLO DIDATTICO	164	180	344	47.67	52.33

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
		L.TEMPE					
12	109	XII CIRCOLO DIDATTICO L.TEMPE	242	172	414	58.45	41.55
12	110	XII CIRCOLO DIDATTICO L.TEMPE	189	204	393	48.09	51.91
12	117	XII CIRCOLO DIDATTICO L.TEMPE	228	187	415	54.94	45.06
12	118	XII CIRCOLO DIDATTICO L.TEMPE	222	180	402	55.22	44.78
12	122	XII CIRCOLO DIDATTICO L.TEMPE	225	181	406	55.42	44.58
12	187	XII CIRCOLO DIDATTICO L.TEMPE	159	128	287	55.40	44.60
13	86	SCUOLA MEDIA C.COLOMBO	271	191	462	58.66	41.34
13	87	SCUOLA MEDIA C.COLOMBO	94	102	196	47.96	52.04
13	119	SCUOLA MEDIA C.COLOMBO	75	62	137	54.74	45.26
13	120	SCUOLA MEDIA C.COLOMBO	206	114	320	64.38	35.62
13	121	SCUOLA MEDIA C.COLOMBO	204	182	386	52.85	47.15
13	123	SCUOLA MEDIA C.COLOMBO	210	163	373	56.30	43.70
13	124	SCUOLA MEDIA C.COLOMBO	230	181	411	55.96	44.04
13	125	SCUOLA MEDIA C.COLOMBO	178	181	359	49.58	50.42
13	126	SCUOLA MEDIA C.COLOMBO	193	175	368	52.45	47.55
13	127	SCUOLA MEDIA C.COLOMBO	278	200	478	58.16	41.84
13	128	SCUOLA MEDIA C.COLOMBO	257	184	441	58.28	41.72
13	129	SCUOLA MEDIA C.COLOMBO	181	125	306	59.15	40.85
13	130	SCUOLA MEDIA C.COLOMBO	163	128	291	56.01	43.99
13	131	SCUOLA MEDIA C.COLOMBO	247	160	407	60.69	39.31
14	1	IST. XXV LUGLIO - G.BETTOLO	171	150	321	53.27	46.73

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
14	132	IST. XXV LUGLIO - G.BETTOLO	241	226	467	51.61	48.39
14	133	IST. XXV LUGLIO - G.BETTOLO	192	190	382	50.26	49.74
14	134	IST. XXV LUGLIO - G.BETTOLO	253	197	450	56.22	43.78
15	135	IST. XXV LUGLIO - G.BETTOLO	239	191	430	55.58	44.42
15	136	IST. XXV LUGLIO - G.BETTOLO	192	222	414	46.38	53.62
15	137	IST. XXV LUGLIO - G.BETTOLO	226	206	432	52.31	47.69
15	138	IST. XXV LUGLIO - G.BETTOLO	184	208	392	46.94	53.06
15	139	IST. XXV LUGLIO - G.BETTOLO	210	170	380	55.26	44.74
15	140	IST. XXV LUGLIO - G.BETTOLO	170	177	347	48.99	51.01
15	141	IST. XXV LUGLIO - G.BETTOLO	181	138	319	56.74	43.26
15	142	IST. XXV LUGLIO - G.BETTOLO	144	141	285	50.53	49.47
15	143	IST. XXV LUGLIO - G.BETTOLO	216	175	391	55.24	44.76
15	147	IST. XXV LUGLIO - G.BETTOLO	233	160	393	59.29	40.71
15	148	IST. XXV LUGLIO - G.BETTOLO	182	151	333	54.65	45.35
16	144	LICEO GINNasio ARISTOSSENSO	111	99	210	52.86	47.14
16	145	LICEO GINNasio ARISTOSSENSO	161	140	301	53.49	46.51
16	146	LICEO GINNasio ARISTOSSENSO	110	112	222	49.55	50.45
16	153	LICEO GINNasio ARISTOSSENSO	169	106	275	61.45	38.55
16	154	LICEO GINNasio ARISTOSSENSO	138	108	246	56.10	43.90
17	155	IST. GALILEO GALILEI	150	90	240	62.50	37.50
17	156	IST. GALILEO GALILEI	98	141	239	41.00	59.00
17	157	IST. GALILEO GALILEI	126	110	236	53.39	46.61
18	158	SCUOLA ELEMENTARE G.B. VICO	106	154	260	40.77	59.23

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
18	159	SCUOLA ELEMENTARE G.B. VICO	154	145	299	51.51	48.49
18	160	SCUOLA ELEMENTARE G.B. VICO	170	171	341	49.85	50.15
18	161	SCUOLA ELEMENTARE G.B. VICO	186	174	360	51.67	48.33
18	162	SCUOLA ELEMENTARE G.B. VICO	122	133	255	47.84	52.16
18	163	SCUOLA ELEMENTARE G.B. VICO	136	126	262	51.91	48.09
18	164	SCUOLA ELEMENTARE G.B. VICO	88	103	191	46.07	53.93
18	165	SCUOLA ELEMENTARE G.B. VICO	178	119	297	59.93	40.07
18	166	SCUOLA ELEMENTARE G.B. VICO	144	200	344	41.86	58.14
18	167	SCUOLA ELEMENTARE G.B. VICO	121	115	236	51.27	48.73
18	168	SCUOLA ELEMENTARE G.B. VICO	150	127	277	54.15	45.85
18	169	SCUOLA ELEMENTARE G.B. VICO	154	177	331	46.53	53.47
18	170	SCUOLA ELEMENTARE G.B. VICO	148	167	315	46.98	53.02
18	171	SCUOLA ELEMENTARE G.B. VICO	122	145	267	45.69	54.31
18	172	SCUOLA ELEMENTARE G.B. VICO	129	116	245	52.65	47.35
18	173	SCUOLA ELEMENTARE G.B. VICO	168	135	303	55.45	44.55
19	174	ISTITUTO COMPRENSIVO L.PIRAND	221	172	393	56.23	43.77
19	175	ISTITUTO COMPRENSIVO L.PIRAND	291	241	532	54.70	45.30
19	176	ISTITUTO COMPRENSIVO L.PIRAND	450	389	839	53.64	46.36
19	177	ISTITUTO COMPRENSIVO L.PIRAND	278	211	489	56.85	43.15
19	178	ISTITUTO COMPRENSIVO L.PIRAND	145	132	277	52.35	47.65
19	179	ISTITUTO COMPRENSIVO L.PIRAND	284	204	488	58.20	41.80
19	180	ISTITUTO COMPRENSIVO L.PIRAND	227	152	379	59.89	40.11

Place ID	Section	Polling place	Bitetti	Tacente	Total	B %	T %
19	181	ISTITUTO COMPRENSIVO L.PIRAND	161	127	288	55.90	44.10
19	182	ISTITUTO COMPRENSIVO L.PIRAND	171	145	316	54.11	45.89
19	183	ISTITUTO COMPRENSIVO L.PIRAND	224	217	441	50.79	49.21
19	184	ISTITUTO COMPRENSIVO L.PIRAND	331	204	535	61.87	38.13
19	185	ISTITUTO COMPRENSIVO L.PIRAND	131	110	241	54.36	45.64
19	186	ISTITUTO COMPRENSIVO L.PIRAND	125	130	255	49.02	50.98

10. Data-quality audit

Issue	Finding	Required action
Workbook structure	Hybrid summary / analysis format rather than normalised database	Transform before publication
First-round denominator	Four highlighted candidate rows sum to the worksheet TOTAL in the working file	Confirm whether other candidates / invalid ballots are excluded
Runoff aggregate	Working section sum differs slightly from published city total cited in narrative sources	Reconcile with official municipal return
Section count	189 section records represented; contextual essay refers to 191 sections	Identify missing, special or non-territorial sections
Addresses	Some records use abbreviated or multi-street descriptions	Create authoritative geocoding table
Macro-areas	Analytical areas are interpretive, not official administrative units	Publish assignment rules and alternatives
Time comparison	2022–2025 comparisons may involve different candidates and political contexts	Separate compositional change from territorial persistence
Social interpretation	Class and occupational readings are currently contextual hypotheses	Validate through census and field evidence

11. Reconciliation protocol

1. Obtain the official section-level files for both rounds and preserve the original download unchanged.
2. Create a source manifest with URL, retrieval date, checksum and election-round metadata.
3. Compare every section number against the municipal section register.
4. Record missing, merged, hospital, prison, overseas or special sections explicitly.
5. Recalculate candidate totals and valid-vote denominators from section-level data.
6. Compare workbook totals with official municipal totals and document every discrepancy.
7. Freeze a clean versioned dataset before producing final maps or claims.
8. Retain the working workbook as a historical input, not as the definitive analytical database.

12. Spatial-data protocol

Step	Requirement
Geocoding	Use authoritative addresses; store latitude, longitude, provider, date and confidence
Polling-place geometry	Represent each building as a point; do not imply it equals the residential territory of voters
Section geography	Acquire official section polygons if available; otherwise declare proxy method
Neighbourhood assignment	Use published rules and test alternative boundaries
Map projection	Use an appropriate local projected CRS for distance and area calculations
Boundary uncertainty	Flag sections crossing neighbourhood or functional-zone boundaries
Temporal consistency	Version all geometries and record boundary changes between elections
Public atlas	Suppress no data at aggregate level unless legally or ethically required; avoid personal data

Important distinction

A polling place is the location at which ballots are cast, not necessarily the centroid or social identity of the voters assigned to its sections.

13. Statistical analysis plan

Method	Purpose	Minimum requirement
Descriptive mapping	Identify spatial variation in turnout and support	Correct denominators and geocoded units
Local deviation scores	Show candidate over- and under-performance	City or comparable-area baseline
Moran's I / LISA	Test spatial clustering	Section polygons or defensible neighbourhood graph
Spatial regression	Estimate association with contextual variables	Census covariates and diagnostics
Boundary discontinuity	Test a proposed divide such as Viale Magna Grecia	Comparable units near boundary and pre-analysis plan
Temporal transition matrix	Compare stable and changing territories	Harmonised boundaries across elections
Ecological inference models	Estimate aggregate transition patterns cautiously	Explicit assumptions and sensitivity analysis
Qualitative process tracing	Identify mechanisms behind observed patterns	Interviews, archives and institutional evidence

14. Neighbourhood campaign indicators

Indicator	Use	Interpretive caution
Electoral weight	Prioritise areas with a large share of valid votes	Population and turnout are distinct
Competitiveness	Identify persuasion territories	Close result may reflect unusual candidate supply
Turnout / retention gap	Identify mobilisation problems	Do not assume the same individuals abstained
Candidate specialisation	Identify local coalition pockets	Requires citywide baseline and stable denominator
Volatility	Identify areas of change across elections	Boundary and candidate changes can create artificial volatility
Institutional density	Assess local contact infrastructure	Organisation count does not equal legitimacy
Issue exposure	Connect policy problems with territory	Avoid inferring opinions from exposure alone
Field confidence	Record strength of local evidence	Strategic recommendations should reflect uncertainty

Permitted strategic use

Area-level message testing, field-office planning, event locations, volunteer deployment, stakeholder engagement and service-related research. Excluded: individual political profiling, sensitive-trait inference, suppression or manipulative exploitation of vulnerability.

15. Fieldwork and validation instruments

Recommended instruments include: neighbourhood walk audits; semi-structured interviews with institutional and community actors; focus groups with explicit recruitment rules; archival review; service-access mapping; transport and accessibility observation; local-media content analysis; and a structured claim-evidence register.

Every interpretation should be assigned a confidence level: observed, strongly supported, plausible, exploratory or unresolved. Campaign recommendations should state which evidence would change the conclusion.

Fieldwork question	Evidence sought
What defines the area locally?	Resident boundaries, landmarks, mobility and names
Which institutions are trusted?	Repeated references, usage, membership and convening power
Which political issues are territorial?	Service gaps, environmental exposure, housing, work, transport
Why does participation differ?	Access barriers, distrust, campaign presence, demographic turnover
Which messengers are legitimate?	Community reputation and cross-network reach
What is missing from the quantitative map?	Informal networks, recent change, conflict and stigma

16. Reproducibility and versioning

Each project release should include a versioned source manifest, raw-data archive, cleaned dataset, transformation script, geocoding table, analytical notebook or script, exported figures and a claim-evidence register. Public outputs should identify the data freeze date.

Suggested naming convention: GHPI_[territory]_[election-or-topic]_[dataset]_[YYYYMMDD]_vX.Y. Material changes to denominators, geometry or candidate results require a major version increment.

Release item	Public / restricted
Main report	Public
Executive brief	Public
Interactive atlases	Public
Clean aggregate dataset	Public where lawful
Raw official downloads	Public link / archived copy subject to terms
Interview transcripts	Restricted and consent-governed
Campaign strategy notes	Client-confidential
Individual personal data	Not collected for this model

17. Ethical and legal checklist

Before use in an electoral campaign, the commissioning organisation should obtain jurisdiction-specific legal advice and document the lawful basis, purpose, retention and access controls for any data processed. The casogeographics model is designed to function with public aggregate data and contextual research; it does not require voter files or sensitive individual profiles.

Check	Yes / No / N.A.
Only aggregate territorial political data are used	
Data sources and calculations are documented	
No protected characteristics are inferred for individuals	
No vulnerable group is targeted for suppression or intimidation	
Fieldwork has consent and safeguarding procedures	
Campaign recommendations include uncertainty and alternatives	
Access to restricted materials is role-based	
Retention and deletion schedule is defined	
Legal review is complete for the intended jurisdiction	

18. References and linked resources

Main Research Report: The Long Geography of Politics: A Geo-Historical Intelligence Framework for Reading Votes, Institutions and Territorial Behaviour.

Taranto interactive source: <https://www.google.com/maps/d/u/0/viewer?mid=1S4lgqOCBgD3KvQzzxy79k22zm3ZgI4o>

Global laboratory: <https://sites.google.com/cultum.org/europeunited/projects/geographies-of-memory/a-global-paleo-political-laboratory>

Case studies: <https://sites.google.com/cultum.org/europeunited/projects/geographies-of-memory/a-global-paleo-political-laboratory/2-case-studies-and-territories>

Method page: <https://sites.google.com/cultum.org/europeunited/projects/geographies-of-memory/a-global-paleo-political-laboratory/1-the-method-the-tree-applied-to-environmental-determinism>

Website: <https://www.casogeographics.com/>

Academic bibliography and detailed comparative-case references are provided in the Main Research Report. The final production release should add official election-data citations, checksums and a definitive data-freeze statement.